Appendix

Figure 1- Descriptive Statistics

	count	mean	std	min	25%	50%	75%	max
Order_ID	2083.0	3207.428709	746.677463	2000.00	2554.500	3201.000	3849.500	4499.00
Units_Sold	2083.0	1.777244	0.753679	0.00	2.000	2.000	2.000	3.00
MRP	2083.0	6080.541704	1644.409666	2006.63	6110.030	6110.030	6110.030	9995.55
Discount_Applied	2083.0	0.620454	0.212954	0.00	0.615	0.615	0.615	1.25
Revenue	2083.0	233.935300	1535.919467	-5010.39	0.000	0.000	0.000	21586.98
Profit	2083.0	1361.577115	1481.887204	-1199.32	67.005	1368.590	2636.440	3999.21

Figure 2- Profit Distribution by Region (Boxplot)

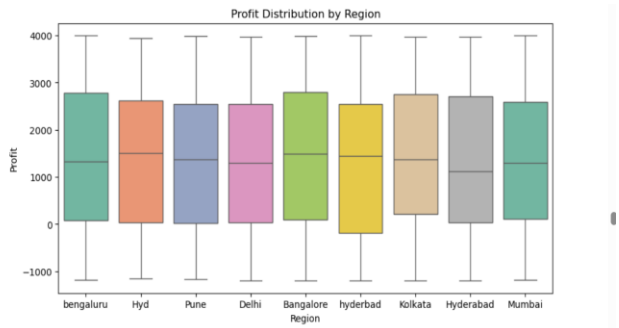

Figure 3- Discount Distribution by Sales Channel Boxplot

Figure 4- Relationship Between MRP and Profit by Sales Channel (Regression Plot)

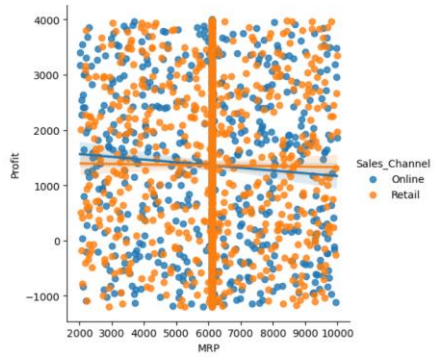

Figure 5-Average Profit by Region

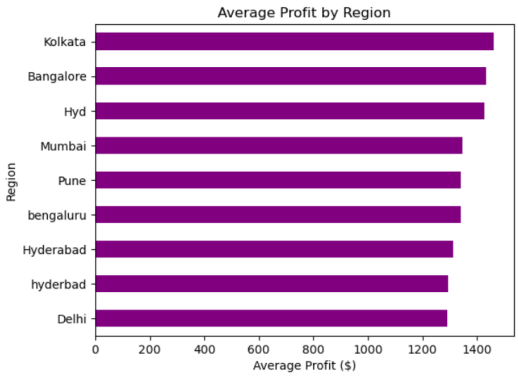

Figure 6a-Product Line Performance

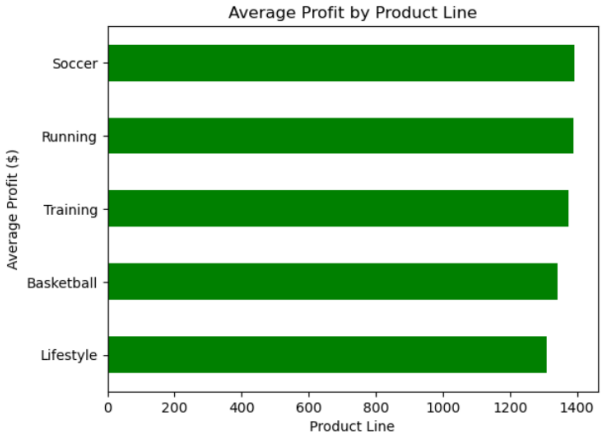

Figure 6b- Profit Share by Product Line

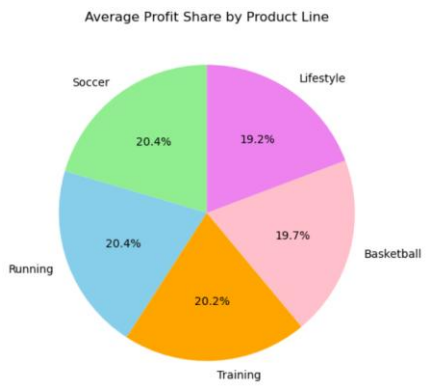

Figure 7- Profit Share by Sales Channel

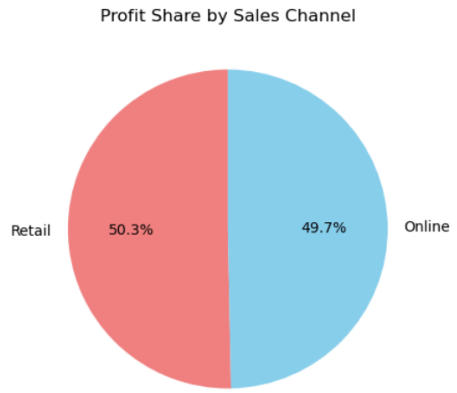

Figure 8- Correlation Heatmap of Discount, Revenue, and Profit

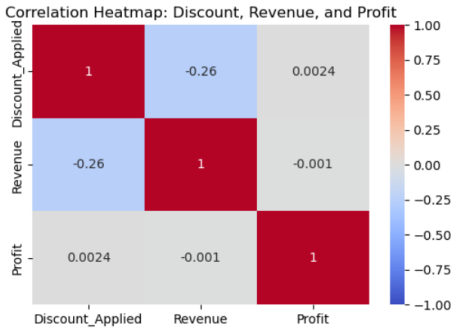

Figure 9- MAE Model Comparison

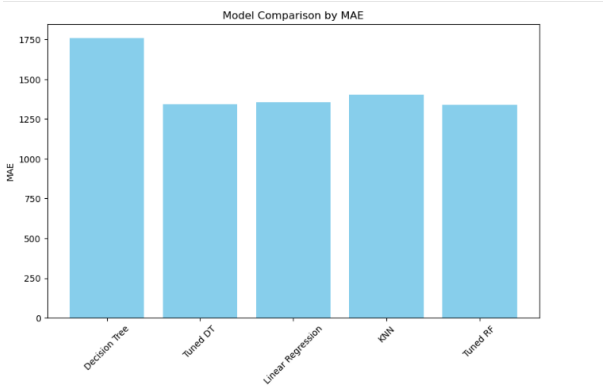

Figure 10-R2 Model Comparison

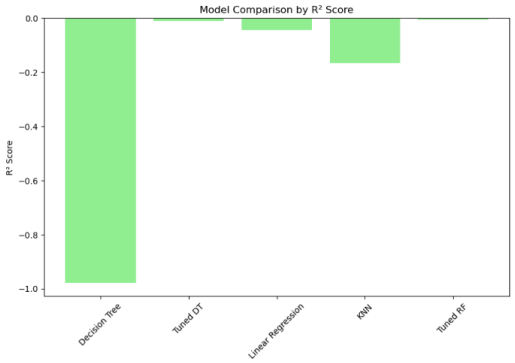

Figure 11-MSE Model Comparison

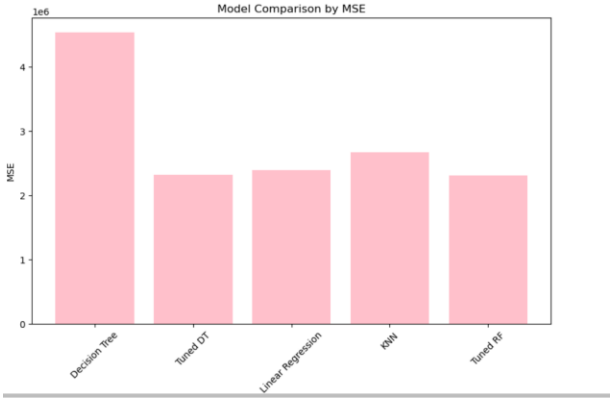

Figure 12-RMSE Model Comparison

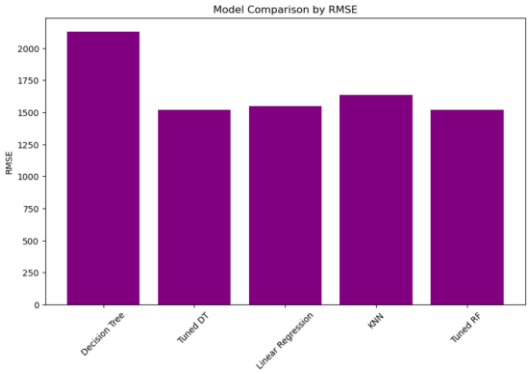
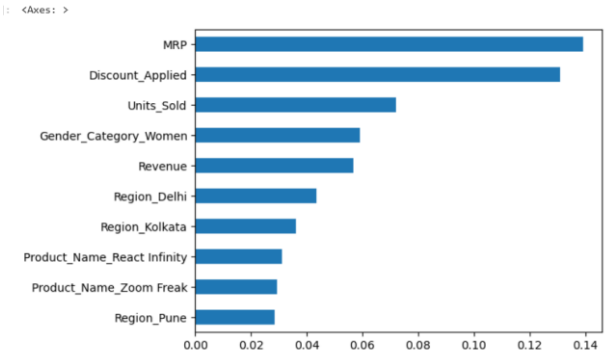
Figure 13- Feature Importances